

Hyunwoo **Park**

AI Researcher · Data Scientist

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"Precision in every data point."

Summary

M.S. student in Applied Artificial Intelligence at Kwangwoon University, advised by Prof. Minsu Cho. My research centers on **label-efficient learning**, i.e., how models can generalize from scarce supervision, with applications in **remote sensing time-series**, **process mining**, and **NLP**. I am also interested in when modern sequence architectures (attention, Transformers) actually pay off outside the large-data regime they were designed for.

Education

Kwangwoon University

Seoul, Korea

M.S. in Applied Artificial Intelligence — *Mar. 2026 – Feb. 2028 (Expected)*

- Advisor: **Prof. Minsu Cho**, DA Lab (Data Analytics Lab)
- Research focus: label-efficient learning (few-shot learning for satellite time-series, federated learning for predictive process monitoring)

Kwangwoon University

Seoul, Korea

B.S. in Data Science, School of Information Convergence — *Mar. 2020 – Feb. 2026*

- **Major GPA: 4.31 / 4.5** (Cumulative GPA: 4.13 / 4.5)

Research Experience

DA Lab (Data Analytics Lab)

Seoul, Korea

Graduate Researcher (M.S. Student) — *Mar. 2026 – Present*

- **Few-shot ecosystem classification from satellite time-series:** classifying 24 **Ecosystem Functional Groups (EFGs)** of the IUCN Global Ecosystem Typology from one-year **HLS (Harmonized Landsat–Sentinel-2)** time-series (24K sites, South Korea) with only 5 labels per unseen class.
- Showed that errors concentrate on same-realm EFG pairs that differ in the *timing* of seasonal events rather than their magnitude, and that global average pooling (GAP) over time erases exactly this timing.
- Proposed a hierarchy-informed pipeline: a **realm-margin loss** that enlarges prototype separation within a realm, and **SegPool**, a training-free inference-time replacement of GAP by normalized calendar-segment pooling, on an **InceptionTime** encoder with transductive α**-TIM** refinement.
- Improved 5-way 5-shot accuracy from **79.0% to 84.4%** over 11 class splits × 8 seeds (positive in all 88 units) and reduced same-realm confusion in every split containing same-realm pairs.
- Conducted a systematic study of **transductive few-shot learning** methods for HLS-based EFG classification; in press at *Expert Systems With Applications* (SCIE, 2026, **second author**), with a preliminary benchmark presented at KIIE 2026 (see Publications).
- **Federated learning for predictive process monitoring:** predicting ED-to-inpatient admission from emergency department event logs (**MIMICEL v2.1.0**, ∼425K cases) without centralizing patient data.
- Compared **FedAvg / FedProx / Ditto / SCAFFOLD** under realistic label skew across sites; FedProx's advantage grows monotonically with heterogeneity.
- Continuing research on dual-process modeling of online review helpfulness (see Publications).

DA Lab (Data Analytics Lab)

Seoul, Korea

Undergraduate Research Intern — *Sep. 2024 – Feb. 2026*

- Studied the cognitive mechanisms of online review helpfulness across four online markets (Amazon, Booking.com, Audible, Coursera) using **400K+ reviews**.
- Proposed a predictive framework based on **Dual-Process Theory**, distinguishing systematic cues (e.g., depth, breadth) from heuristic cues (e.g., rating deviation, recency).
- Built NLP feature pipelines: **RoBERTa**-based representations for emotional arousal and **NMF (Non-negative Matrix Factorization)** for topic breadth.
- Applied **PLS-SEM** for latent construct validation and developed a **multilevel ZINB (Zero-Inflated Negative Binomial)** model to handle zero-inflated counts and item-level heterogeneity.
- Key findings: (1) systematic cues consistently drive helpfulness across all domains, whereas (2) heuristic cues have stronger influence on experience-goods platforms (e.g., Audible).
- Resulted in a co-authored paper in *IEEE Access* (SCIE, 2026) and a first-author paper under review at *Electronic Commerce Research* (Springer).

Publications

Journal Articles

Transductive Few-shot Learning for Ecosystem Classification from Satellite Time Series: A Systematic Study — *Expert Systems With Applications (SCIE)*

Second Author (In Press) — *2026*

Minsu Cho, **Hyunwoo Park**, Dohyun Park, Gyunam Park, Seonyoung Park. *Expert Systems With Applications*, 134761, 2026 (in press). [DOI]

When Does Systematic Processing Add Value? A Dual-Process Analysis of Review Helpfulness across Online Markets — *Electronic Commerce Research*

First Author (Under Review)

Hyunwoo Park, Sujin Jung, Heeryeong Park, Minsu Cho. *Electronic Commerce Research* (Springer).

Information Processing and Review Helpfulness in Online Platforms: A Dual-Process Perspective — *IEEE Access (SCIE)*

Co-Author — *2026*

Heeryeong Park, Pilkyu Chung, **Hyunwoo Park**, Sujin Jung, Minsu Cho. *IEEE Access*, 2026.

Conference Papers (Domestic)

Comparing Transductive Few-Shot Learning Methods for HLS Time-Series Based EFG Classification — *KIIE Spring 2026*

First Author — *2026*

Hyunwoo Park, Dohyun Park, Minsu Cho. *Proceedings of the 2026 Korean Institute of Industrial Engineers (KIIE) Spring Joint Conference*, pp. 641–644.

Design and Implementation of 'Lec-Q': A Two-Way Learning Support Service to Enhance Understanding of Real-Time Online Lectures — *HCI Korea 2026*

First Author — *2026*

Hyunwoo Park, SeongYun Cheon, AhReum Yoo, Kyudong Park. *Proceedings of HCI Korea 2026*.

Proposal of Neuro-Symbolic AI-based Dark Pattern Detection and User Assistance System in Elderly Kiosk Environments: Focusing on Human-Robot Interaction — *ICROS 2025*

Co-Author — *2025*

Sujin Jung, Soyoung Park, **Hyunwoo Park**, MinJin Oh, Minsu Cho. *Proceedings of the Institute of Control, Robotics and Systems (ICROS) Annual Conference 2025*, pp. 958–959.

Development of 'PHOTATO', a Photo Analysis Travel Recommendation System — *HCI Korea 2025*

Co-Author — *2025*

HaKyeong Lee, HanBee Ko, EunJin Joo, **Hyunwoo Park**, SeongYun Cheon, SeoYeon Jeon, YongMin Kim, Wonjoon Kim. *Proceedings of HCI Korea 2025*, pp. 1211–1215.

Projects

Lec-Q: AI-based Interactive Learning Platform — *Seoul, Korea*

Project Lead & Main Developer (First Author) — *Sep. 2025 – Dec. 2025*

- Developed a real-time interactive learning platform using **WebRTC** to close communication gaps in online lectures.
- Implemented a **RAG (Retrieval-Augmented Generation)**-based Q&A chatbot (OpenAI API) that answers from lecture materials, with a two-step instructor verification.
- Designed real-time handwriting sharing that automatically captures and distributes the instructor's notes to reduce students' cognitive load.
- Built an instructor analytics dashboard visualizing time-stamped question distribution and student concentration levels.
- Stack: **React, Node.js, WebRTC, OpenAI API**. Published as **first author** at **HCI Korea 2026**.

Neuro-Symbolic AI-based Dark Pattern Detection Service — *Seoul, Korea*

Full-Stack Developer & AI Researcher (Graduation Project) — *Jan. 2025 – Sep. 2025*

- Detected deceptive designs (dark patterns) on websites and kiosks using **Neuro-Symbolic AI**, combining BERT-based deep learning with logical reasoning for high precision.
- Delivered a **Chrome Extension & Web Dashboard**; led full-stack development with **React, Node.js, and Flask**, including AI model serving and **MongoDB** integration.
- Extended the work to offline kiosk environments to assist elderly users, with a focus on Human-Robot Interaction (HRI).
- Won the **Encouragement Award (3rd Place)** at the Graduation Project Exhibition (Sep. 2025).
- Co-authored a paper at **ICROS 2025** (HRI track).

PHOTATO: Photo Analysis-based Personalized Travel Recommendation System — *Seoul, Korea*

Team Lead & Backend Developer (Co-Author) — *Nov. 2024 – Feb. 2025*

- Led development of a travel recommendation service that suggests destinations from visual features of user-uploaded photos using **CNNs**.
- Engineered backend route optimization using a **TSP (Traveling Salesperson Problem)** algorithm to generate efficient travel paths.
- Conducted user research (surveys, in-depth interviews) and derived UX insights with personas and journey maps to define core features.
- Co-authored a paper in **Proceedings of HCI Korea 2025**.

Honors & Awards

Awards

Sep. 2025 **Encouragement Award (3rd Place),** Graduation Project Exhibition (Dark Pattern Detection Service) *Seoul, Korea*

Dec. 2024 **Grand Prize (Poster Session),** MATCH-UP Competition *Seoul, Korea*

Dec. 2024 **Grand Prize (Practical Report),** MATCH-UP Competition *Seoul, Korea*

Scholarships

Mar. 2025 **Academic Excellence Scholarship,** Kwangwoon University *Seoul, Korea*

Sep. 2024 **Full Tuition Scholarship (1st Class),** Kwangwoon University *Seoul, Korea*

Sep. 2024 **Academic Excellence Scholarship,** Kwangwoon University *Seoul, Korea*

Sep. 2024 **Dean's List,** Kwangwoon University *Seoul, Korea*

Sep. 2023 **Academic Excellence Scholarship,** Kwangwoon University *Seoul, Korea*

Teaching Experience

D.Lab Code Academy *Seoul, Korea*

Coding Instructor *Dec. 2024 – May 2025*

- Taught fundamental coding concepts and computational thinking to elementary school students.
- Led hands-on programming activities to build problem-solving and logical reasoning skills in young learners.

Kwangwoon Artificial Intelligence High School *Seoul, Korea*

Lead Instructor & Curriculum Developer *Jan. 2024 – Nov. 2024*

- Co-designed the syllabus and teaching materials for a special Python course with the school's teachers.
- Taught the 1st-semester course (10 sessions, Mar.–Jul. 2024) on Python fundamentals (control flow, functions, data structures) and data analysis libraries (NumPy, Pandas, Matplotlib).
- Re-selected as instructor for the 2nd semester based on strong teaching evaluations and student feedback.
- Taught the 2nd-semester course (10 sessions, Sep.–Nov. 2024) on web crawling and big-data analysis (text mining, geospatial analysis, sentiment analysis, recommender systems), and supervised final team projects.

Extracurricular Activity

CHIC (Data Science & AI Academic Club) *Seoul, Korea*

Core Member & Academic Mentor *Sep. 2023 – Feb. 2026*

- Mentored 2nd- and 3rd-year Data Science majors on core coursework and problem-solving (Mar.–Jun. 2025).
- Ran intensive Deep Learning & AI study sessions covering theoretical foundations and model architectures (Dec. 2023 – Feb. 2024).
- Participated in regular academic sessions on Python programming, data analysis, and machine learning algorithms.

Sowon (SW Volunteer Corps) *Seoul, Korea*

Curriculum Division Team Leader & Lead Instructor *Sep. 2023 – Feb. 2025*

- Served as Lead Instructor at Kwangwoon Artificial Intelligence High School for one year (see Teaching Experience).
- Mentored student projects at the 'And Center' Youth Hackathon (Nov. 2023, Nov. 2024).
- Volunteered as a coding instructor at Seoul Jeongmin Special School (Dec. 2023 – Mar. 2024).

Student Council *Seoul, Korea*

Student Council Member & Department Representative *Mar. 2024 – Feb. 2026*

- Represented the department at Freshman Orientation, presenting the academic curriculum and career paths (Feb. 2025).
- Ran consultation and Q&A sessions for undeclared students on major selection and educational roadmaps.

Skills

Machine Learning	Few-shot learning (ProtoNet, transductive α-TIM), federated learning (FedAvg, FedProx, Ditto, SCAFFOLD), time-series encoders (InceptionTime), temporal pooling, BERT / RoBERTa, CNN, RAG
Statistics	PLS-SEM, multilevel zero-inflated count models (ZINB), NMF topic modeling
ML Tools	Python, PyTorch, scikit-learn, Hugging Face, Pandas, NumPy
Geospatial	Google Earth Engine, HLS (Harmonized Landsat–Sentinel-2)
Web & Backend	React, Node.js, Flask, MongoDB, WebRTC
Tools	Git, Linux, LaTeX, Google Colab